

Math 445/545. Winter 2026. Practice problems for the Midterm 1.

The problems below illustrate various techniques you are supposed to know for the final. Some of the problems below are more difficult than the ones you will see on the midterm.

Problem 1. Let a be an element of a ring R . Let $S = \{x \in R \mid ax = 0\}$. Show that S is a subring of R .

Problem 2. Assume that a is a nilpotent element of a unital ring R . Prove that $1 + a$ is a unit of R .

Problem 3. What is characteristic of the ring $\mathbb{Z}_4 \times 4\mathbb{Z}$?

Problem 4. What is the number of irreducible polynomials of degree 3 in $\mathbb{Z}_2[x]$?

Problem 5. Compute evaluation $\phi_2(x^{100} + 2x^{31} + 3x^2 + 5)$ when the polynomial is considered over $\mathbb{Z}_{11}$ (so the answer should be some element of $\mathbb{Z}_{11}$).

Problem 6. In the ring $\mathbb{Z}_{3600}$ find the number of idempotents, nilpotents, units, and zero divisors.

Problem 7. What is the remainder of $x^{100} + 1 \in \mathbb{R}[x]$ when divided by $x^2 - 2x$?

Problem 8. Suppose that R is a ring such that $x^3 = x$ for all $x \in R$. Prove that $6x = 0$ for all $x \in R$.

Problem 9. (a) Prove that the ring $\mathbb{Z}[\sqrt{2}]$ has infinitely many units.

(b) Prove that the ring $\mathbb{Z}[i]$ has only finitely many units and find them all.

Problem 10. Is the group $\mathbb{Z}_{143}^\times$ cyclic? ($143 = 11 \cdot 13$).